



Reimagining Claims Processing with Agentic AI – Powered by EMA

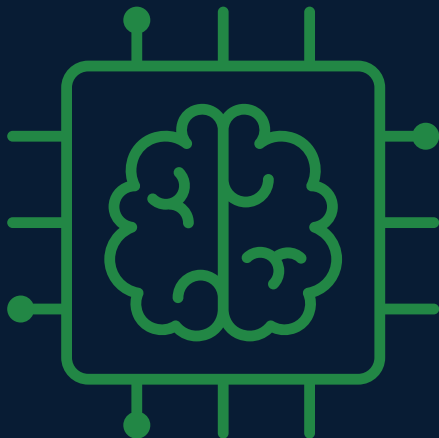
Designing the next-generation claims
automation platform

By Pratik Patil

Understanding the Claims Journey: Critical, High-Pressure Process

Vehicle insurance claims are emotionally and financially sensitive moments for customers. After an accident, policyholders want immediate clarity:

- what should i do now?
- who will guide me?
- How long will the claim take?
- Will I actually get the payout?
- whom should i contact first?
- Do I need to arrange repairs myself?
- How much will I be out of pocket?



Even though insurance companies have established workflows, the process often feels slow, fragmented, and confusing to customers and internal teams.

Whats broken for coustomers and why it matters?:

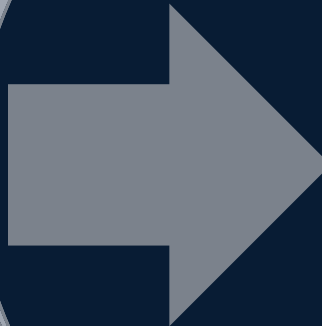
Although insurers have defined processes and teams, the current claims journey still fails customers. Below are the core consumer-facing problems each line names who is involved and the immediate impact.

- Unclear next steps & timelines (Claimant → Intake Specialist → Adjuster)
- Customers don't know what happens after reporting a claim → leads to repeated follow-ups & anxiety.
- High information burden & repeated evidence requests (Claimant → Intake → Adjuster → Appraiser)
- Slow, inconsistent damage assessment and coordination (Adjuster ↔ Appraiser / Vendor)
- Opaque approvals & payout uncertainty (Adjuster → Manager → Payout) — plus noisy SIU flags

First Notice of Loss (FNOL)



Triage & Assignment



Investigation & Estimation



Settlement & Payout / SIU

- Customer reports the accident + uploads photos
- Intake specialist clarifies missing details manually

- Claim is manually reviewed and categorised
- Often misassigned, causing delays and rework

- Adjuster checks photos, documents & repair reports
- Damage estimate and decisions take time due to scattered info

- Approvals, verification & payouts handled manually
- Fraud cases flagged inconsistently delays for customers

Our MVP Proposal: Introducing Agentic AI for Claims Processing

Customers face stress and delays due to repeated handoffs and manual checks across intake, adjusters, appraisers, and SIU. An agent-driven system streamlines this by clarifying next steps, reducing manual workload, standardizing estimates, and making approvals fast, predictable, and transparent.

What This Enables?:

- Faster, more accurate claims decisions
- Reduced operational load
- Fewer customer escalations
- Higher trust and transparency
- A consistent experience across all claims

How Agentic AI Improves the Claims Experience?:

- Automates intake work by extracting structured data from calls, photos, and user descriptions.
- Gives instant clarity with quick incident summaries, damage cues, and predicted next steps.
- Standardizes routing by analyzing severity, coverage, and fraud signals to assign the right workflow.
- Supports adjusters with unified insights from photos, docs, estimates, and liability indicators.
- Speeds up resolutions by coordinating repairs, requesting missing docs, tracking dependencies, and giving transparent timelines.

KPI'S BOX:

- Time-to-first-response ↓
- Auto-route low-complexity claims
- SIU false positives ↓

AI Agents to solve the problems of vehicle Insurance claims?

FNOL Agent-AI powered Claims intake assistant

- Greets user, verifies policy, and starts structured FNOL flow
- Asks smart, context-based questions to capture what/where/when
- Collects photos/docs and extracts key incident details automatically
- Generates a clean incident summary + next steps for adjuster
- Includes a single human-in-loop step for clarification when needed



Payout Orchestration Agent

- Auto-verifies documents
- Confirms policy coverage
- Calculates payable amount
- Applies limits & deductibles
- Checks fraud/compliance flags
- Ensures eligibility validation
- Requests missing documents
- Coordinates approvals workflow
- Notifies managers/adjusters
- Triggers payout via API
- Generates settlement summary
- Sends summary to customer

Accident Severity & Liability Agent (Photo + Context Estimator)

- Accepts FNOL summary, multi-angle photos/videos, metadata (GPS, time, vehicle), and police report.
- Uses multimodal models to classify severity and detect damaged parts + scene cues.
- Applies liability reasoning (LLM + policy rules) to estimate likelihood of policyholder fault.
- Generates a decision packet: severity, repair-cost band, confidence, key evidence, next step.
- Uses rule thresholds to decide routing:
- Low-severity + high confidence → straight-through processing.
- Medium confidence → adjuster review (HITL).
- High-risk / low confidence → senior adjuster or SIU.
- Orchestrates follow-ups: schedule appraiser, notify vendor, or call third-party specialists automatically.
- Provides full explainability via saliency maps + textual rationale for audit & regulatory compliance.

ESTIMATOR AGENT -End To End



- Intakes multimodal FNOL summary, photos/videos (multi-angle), past repair docs, vehicle details, odometer, GPS/time metadata, and local labor/parts pricing.
- Detects & localizes damage across components (bumper, door, chassis, lights, glass, frame).
- Identifiy severity: surface dent, panel replacement,Damage



- Produces a full repair workplan: itemized parts, labor hours, vendor rates, and estimated cost range.
- Generates evidence-backed rationale with saliency maps, annotated photos, and short textual justification.
- Recommends next action: repair scheduling, further inspection, total-loss evaluation.



- Suggests optimal vendors based on part availability, distance, lead times, and policy rules.
- Creates a structured adjuster packet: confidence, itemization, rationale, and recommended next steps (auto-schedule / HITL).
- Logs outcomes (overrides, invoices, repair completion) for retraining, audit, and future model improvement.



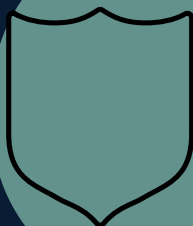
FRAUD DETECTION & RESPONSE AGENT



- Analyzes claimant behavior: repeat claims, unusual patterns, prior fraud flags, and cross-claim activity.
- Evaluates FNOL text, photos, metadata & documents to detect inconsistencies, metadata mismatches, reused images, time/location anomalies, and invoice irregularities.



- Assigns a fraud risk score with clear explanations (“repeat claimant,” “metadata mismatch,” “damage deviation”).
- Uses risk-based decision logic: requests extra documents, clarification calls, in-person verification, payout pause, or fast-track clearance.
- Medium-risk cases require human approval; **REDUCE FP**



- Automatically escalates high-risk cases to adjusters or SIU with a compact evidence packet and recommended actions.
- Maintains full audit trails — logging risk factors, decisions, overrides, outcomes.
- Continuous learning loop improves accuracy and compliance over time.

Overview		Our MVP Proposal			Prioritization & Rationale		Live demo overview	
Feature No.	Feature Name	Reach	Impact	Confidence	Effort	RICE Score	Priority	
1	FNOL Agent (AI-Assisted First Notice of Loss)	5 Affects all claimants; highest usage	5 Reduces confusion, improves data quality	4 LLM intake is stable	2 Moderate build	50	1	
3	Estimator / Appraiser Automation Agent	4 Applies to most damage claims	5 Directly reduces adjuster workload; huge time savings	4 High feasibility with image datasets + historical repair costs	3 Moderate-high but predictable	26.6	2	
2	Accident Severity & Liability Agent	5 Used across all accident claims	4 Speeds routing but still needs HITL	3 Liability decisions are tricky; uncertainty higher	3 Moderate effort	20	3	
4	Fraud Detection & Response Agent	3 Small segment of claims	5 High financial protection	3 Needs historical fraud patterns	4 High complexity	11.25	4	
5	Payment Orchestration & Payout Agent	5 Affects all approved claims	3 Improves payout experience	2 Backend integration risk	4 Integration-heavy	7.5	5	

Feature Description FNOL AGENT:

- 1. Guided Claim Submission Simple step-by-step chatbot flow that collects all required FNOL details.
- 2. Automated Data Capture Conversation turns into structured claim data without manual entry.
- 3. Auto-Generated Incident Summary System creates a clean adjuster friendly summary instantly.
- 4. Real-Time Claim Tracking Users see every stage: Submitted → Review → Assessment → Payment.
- 5. Easy Document & Photo Uploads Upload damage photos, reports, and proofs directly in flow.
- 6. Reduced Agent Workload Automation removes repetitive 20-minute intake calls.
- 7. Faster Claim Resolution Clean data + complete documents = quicker decisions and payouts.

Feature prioritization:

- Highest Usage Workflow
- Biggest Customer Pain Point
- High Operational Cost Reduction
- Improves Data Quality,Measurable ROI

WORKING FLOW FOR NEW CUSTOMER & THEIR INCIDENT

File New Claim

Step 1 of 3

Policy Information

Incident Details

Contact Information

Policy Number

POL-123456

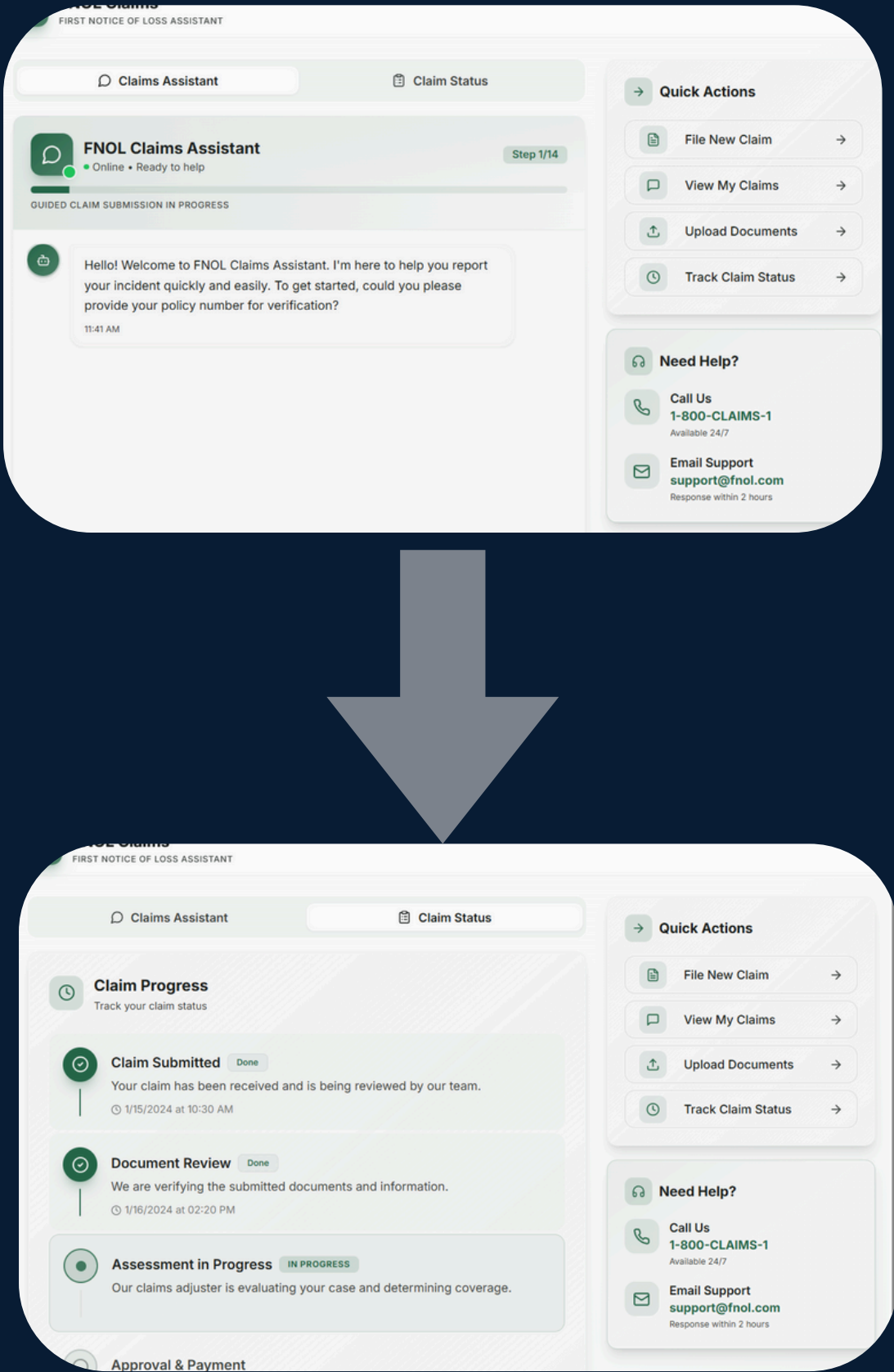
Incident Date

dd-mm-yyyy

Incident Type

Select incident type

WORKING FLOW FOR EXISTING CUSTOMER

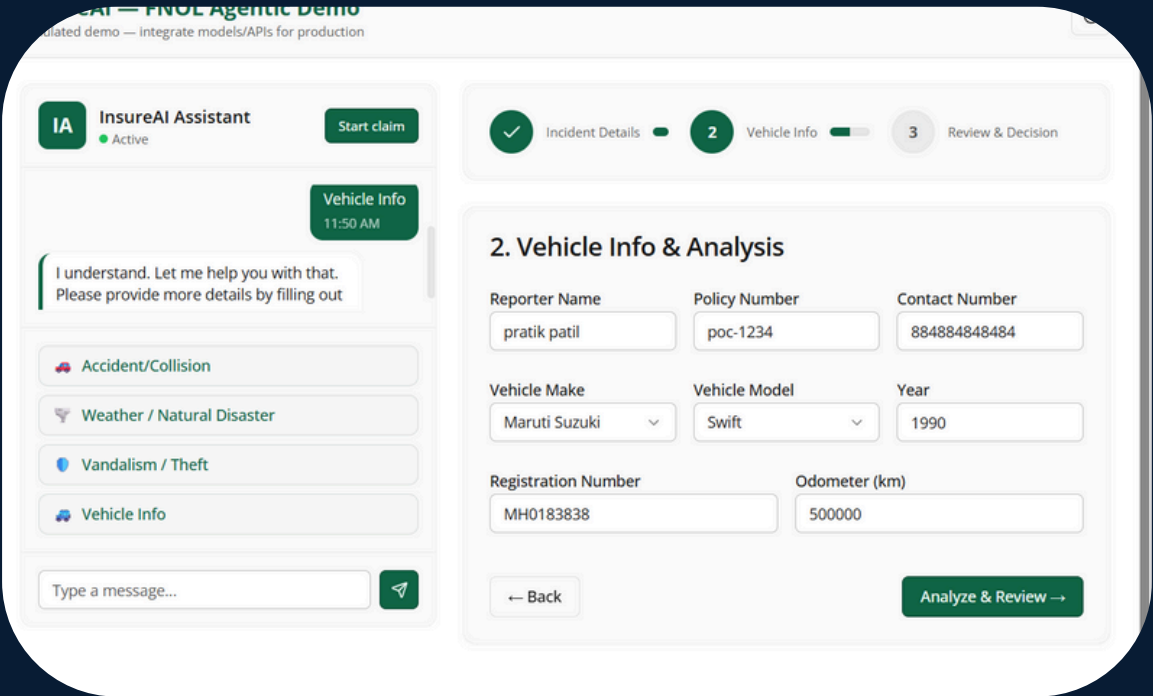
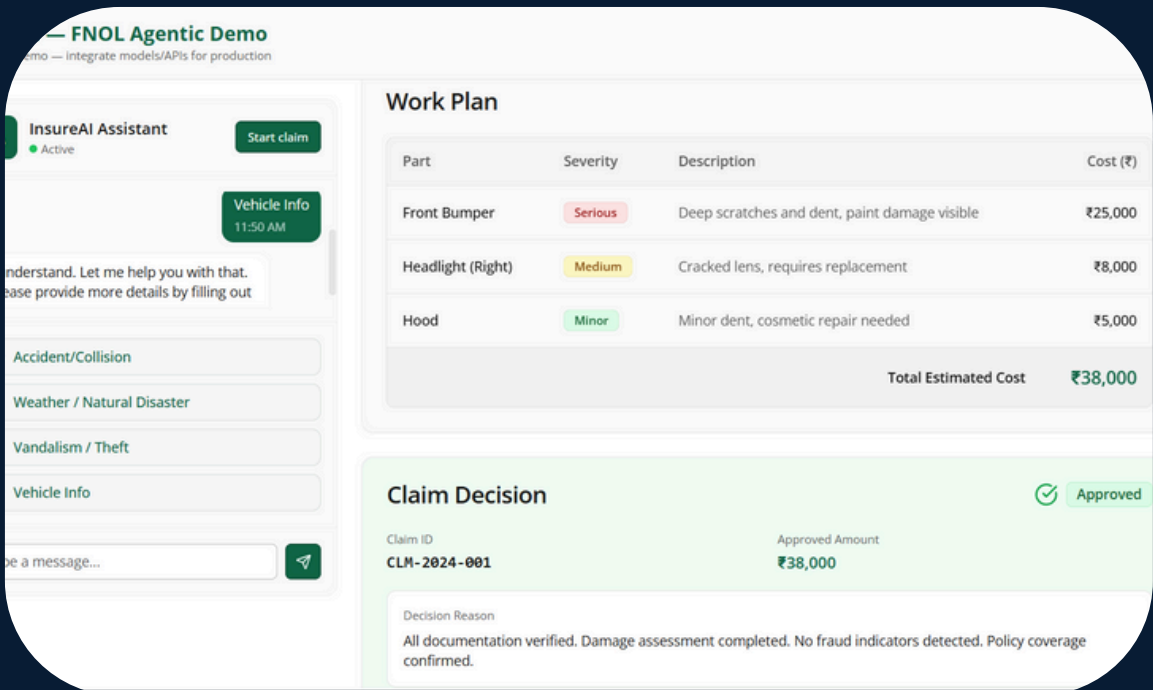
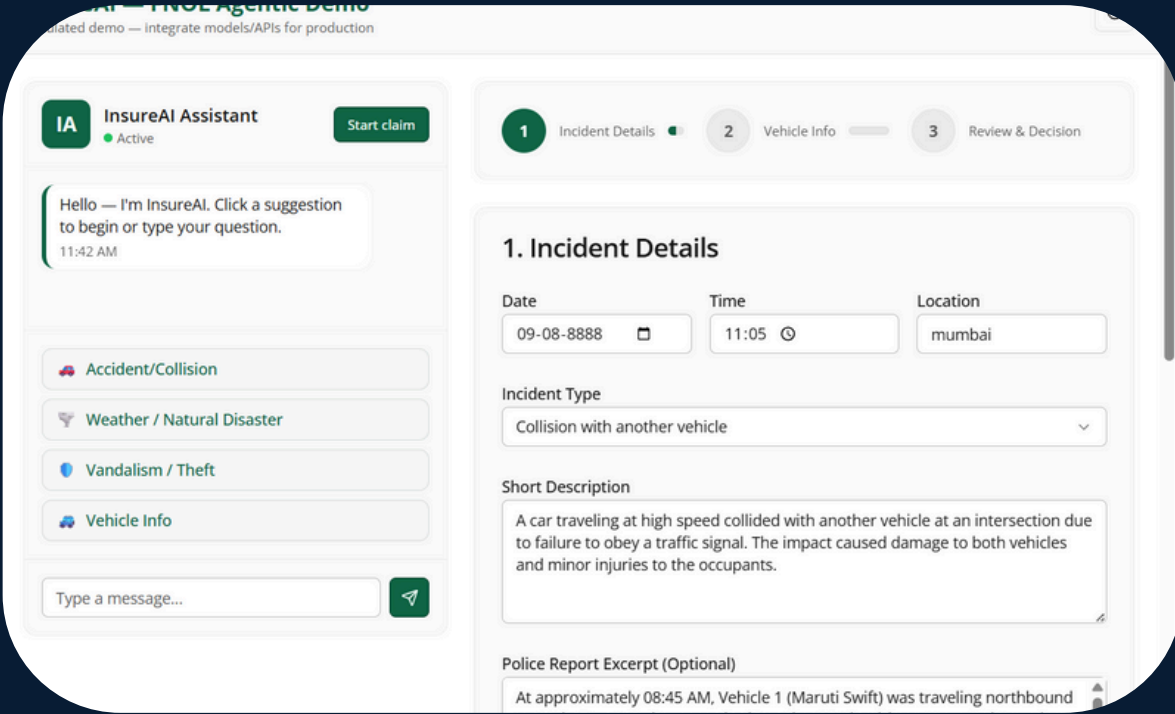


WORKFLOW FOR ESTIMATOR AGENT

KEY FEATURES:

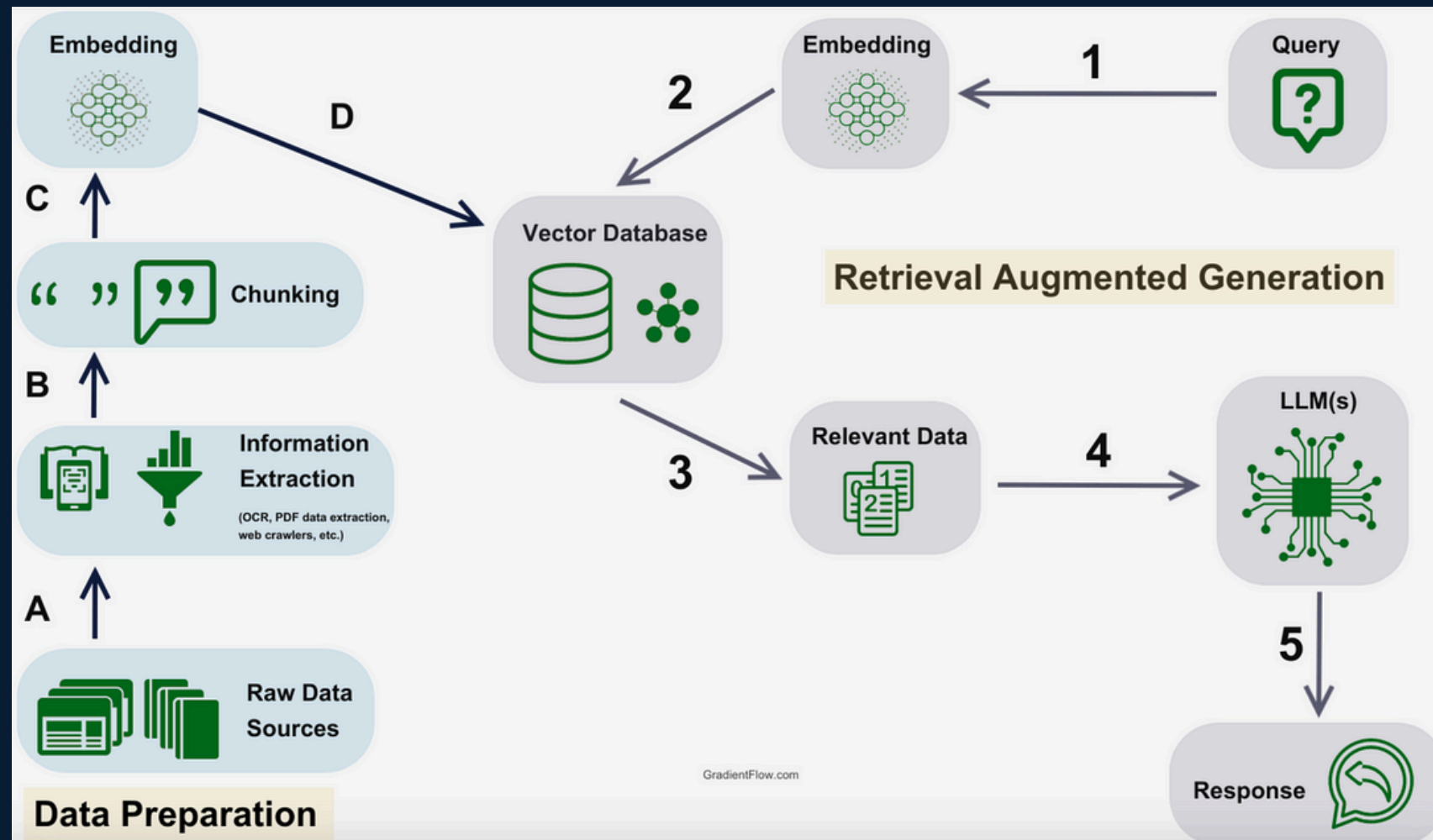
- AI damage detection
- Multi-angle photo analysis
- Auto repair cost estimation
- Parts & labor pricing engine
- Replace vs. repair decisioning
- Smart repair workplan
- Vendor/garage recommendations
- Instant cost range output
- HITL review for low confidence
- Continuous learning from invoices

- High automation feasibility
- Strong historical data availability
- High adjuster time savings
- Direct customer clarity
- Major cycle-time reduction
- High financial ROI
- Enables straight-through processing



Resilience in Product Development — AI Agent Pipeline (Mantra IAS)

- Faced major reliability issues in the early chatbot version — hallucinations, slow retrieval, and low student trust.
- Took ownership of the full AI agent pipeline: content ingestion → RAG retrieval → agent decisions → user response.
- Iterated relentlessly using a feedback loop: error logs, faculty inputs, and rapid prompt/chunking experiments.
- Added guardrails (“I don’t know”), citations, and reworked retrieval logic to raise answer accuracy by 18–25%.
- Result: Faculty query load dropped 55%, response time dropped from 12 hrs → under 1 min, CSAT reached 4.6/5.

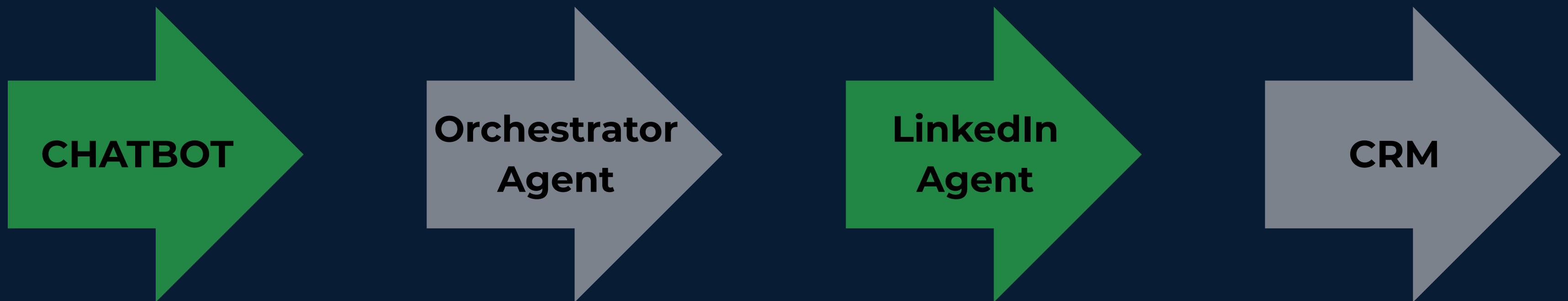


- Multi-agent validation layer
- Adaptive retrieval thresholds
- Auto failover logic
- Agent-level instrumentation
- Synthetic edge-case testing
- Latency-optimized pipeline
- Session-state cleanup

Resilience = continuous iteration despite failure signals, and owning the problem end-to-end until impact is achieved.

Resilience: Automating Marketing With AI Agents + LinkedIn (CCTech).

- Faced messy, unstructured documents and inconsistent lead-generation workflow that required heavy manual effort.
- Re-architected the AI system using LangGraph + AI agents to automate the entire marketing pipeline end-to-end.
- Connected chatbot → LinkedIn outreach agent → CRM; built personalized message sequences and auto follow-ups.
- Overcame LinkedIn rate limits, API failures, and data-quality gaps by building retry logic + metadata cleaning.
- Result: Achieved 25% faster retrieval, and reduced manual marketing work by >50%, improving qualified lead flow.



Resilience = solving ambiguity, fixing broken flows, and shipping a functioning automated system despite constraints.